SRR - Site Risk Register

Purpose and risk categories

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Study / protocol GLENOS-01 / GS-2026-01	**CIP version / date** CIP v4.2 / 12 Jun 2026	**Site / code** Riverside Orthopedic Center / 014	**SRR ID / revision** SRR-014 / r3
Principal Investigator Dr Alex Morgan	**Register owner / role** Maria Silva / Study Coordinator	**Sponsor risk information (ID / version)** GS-RA-01 v2 (sponsor risk assessment)	**Study phase** Enrollment and follow-up
Date opened 2 Jun 2026	**Last review / next review** 31 Oct 2026 / 31 Jan 2027	**Current reviewer / role** Dr R. Teixeira / Sub-investigator	**Controlled location** ISF tab 3 (Quality) / SRR-014

A Purpose and use

The SRR records the risks that this site's own way of working creates for the investigation: gaps in staff, workflow, equipment or experience that could compromise a condition critical to quality (CtQ). The sponsor's risk assessment and the manufacturer's device risk file cannot see these from outside, and the SRR does not copy them. Open it before the first participant is included and keep it current until close-out.

How the form works

- **Pages 1 and 2** are reference: risk categories, CtQ domains and the rating guide.
- **E and F** are the register: one row per risk, with the same Risk ID on both pages. Reprint the pair when it fills.
- **G** records every review of a risk; **H** captures an urgent risk before it enters the register. Each is reprinted on its own.
- **I** holds the signed decisions to lower, accept, close or escalate a risk; **J** closes each review of the whole register.

Unit of analysis	Decision supported	Inputs	Outputs
One local risk to a CtQ factor of this investigation	What threatens the factor, what risk remains with the controls in place, what happens next, who owns it and when it is reviewed again.	CIP, sponsor risk information, local workflow and capacity, and findings from the IMC, PIR, CSST, SLCT, SART and CAPA log.	Controls to verify (IMP) PI decisions (POT) Phase-gate conditions (MRL) CAPA, sponsor and SART routes

B Risk categories

Risk category	Typical local vulnerability	What may be affected
Safety / clinical event	Delayed safety awareness, incomplete assessment or missed reporting.	Participant protection and timely clinical decisions.
Procedure / surgeon performance	Learning-curve drift, unplanned substitution or inconsistent technique.	Procedural consistency and interpretation of outcomes.
Device accountability	Mixed or poorly segregated stock, missing identifiers or reconciliation gaps.	Component selection and participant-device traceability.
Data / endpoint reliability	Missing assessment, unresolved critical query or source-to-CRF mismatch.	Complete, reliable and interpretable study data.
Operational / staffing	No assessor on needed days, turnover, workload or unclear handover.	Capacity to deliver required study activities.
Consent / eligibility	Wrong consent version, delayed re-consent or unsupported eligibility.	Participant rights and valid enrollment.
Imaging / assessment workflow	Wrong projection, missed window or inconsistent assessment method.	Comparable and interpretable endpoint assessments.
Vendor / external data	Delayed external records, inaccessible results or missing transfers.	Availability of safety and endpoint evidence.

C CtQ domains

In the register, give the domain code and write the concrete factor of this investigation that the risk threatens, with its source (CIP, sponsor risk information, local workflow). A domain name is not a factor.

Code	CtQ domain	What it covers
CtQ1	Site capability, readiness and capacity maintenance	Staff, volume, facilities, equipment, imaging and follow-up capacity
CtQ2	Participant selection and baseline characterization	Eligibility criteria, baseline assessment, documented selection decision
CtQ3	Informed consent and participant understanding	Consent process, version control, comprehension, re-consent on amendment
CtQ4	Device management, accountability and traceability	Receipt, storage, use in a named participant, identifiers, reconciliation
CtQ5	Procedural delivery and surgical technique standardization	Approach, positioning, soft tissue, intraoperative decision thresholds
CtQ6	Training, credentialing and learning-curve control	Competency, device release, maintained authorization, learning effects
CtQ7	Perioperative and postoperative care pathway and follow-up retention	Care pathway, visit scheduling and windows, retention
CtQ8	Endpoint framework and outcome-assessment standardization	Assessor consistency, administration conditions, PRO timing and mode
CtQ9	Imaging acquisition, interpretation and endpoint control	Acquisition parameters, operator consistency, local review, timing
CtQ10	Safety surveillance and adverse device event management	Detection, classification, causality and reporting of AEs and DDs
CtQ11	Data capture, source documentation and data-flow integrity	Source adequacy for reconstruction, source-to-record consistency
CtQ12	Regulatory compliance and deviation management	Detection, documentation and correction of deviations

D Risk rating guide

Rate with the controls already in place; a planned action does not lower a score. Write N/A, with a short reason, when a dimension does not apply. Do not average or multiply. A reportable event follows its CIP route at every priority.

Rating field	1	2	3
Probability	Unlikely or isolated vulnerability.	Plausible in normal site conduct.	Likely, recurrent or already observed.
Safety	No expected effect on safety or participant protection.	May affect timely clinical review, protection or safety documentation.	May cause or hide serious harm, a serious device effect or a major safety concern.
Procedure	No expected effect on the conduct of the procedure.	May affect consistency, documentation or protocol adherence.	May affect surgical appropriateness, intraoperative decisions or technique.
Device	No traceability or selection concern.	Recoverable discrepancy or delayed reconciliation.	Missing or conflicting identifiers, wrong component or broken participant-device link.
Data	No expected effect on critical data.	May affect completeness, timeliness or source-to-CRF consistency.	May compromise endpoint interpretation, eligibility evidence or critical data.
Detection	Easy: found by routine checks before it has an effect.	Moderate: found only at a planned review, reconciliation or second check.	Difficult: may stay hidden until endpoint review, inspection or recurrence.

Low	Moderate	High	Critical
All impact dimensions are 1, probability is low and the problem is easy to detect.	At least one dimension is 2, or probability is medium, but no dimension is 3 and controls appear adequate.	Any dimension is 3, or the problem recurs, is poorly controlled, hard to detect or needs active PI review.	Immediate or unresolved safety risk, broken participant-device link, unrecoverable endpoint data, or an event to escalate now.

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Main register · identification and rating

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Study / site	SRR ID / revision	Revision effective date	Register sheet no. / total
GLENOS-01 / 014	SRR-014 / r3	31 Oct 2026	1 / 1

E Main register · identification and rating

Name the domain, the concrete factor threatened and its source; describe the local vulnerability and its consequence; list the controls already in place with their evidence; then rate. Continue the same Risk ID in F.

Risk ID	Date opened	Category	CtQ domain, factor and source	Local vulnerability and possible consequence	Existing controls and evidence	Prob.	Safety	Proc.	Device	Data	Detect.	Priority
SRR-001	2 Jun 2026	Operational / staffing	Domain: CtQ7 Factor: 6-week and 3-month visits done in window with the Constant score complete Source: CIP 9.2	Only one physiotherapist is trained for the Constant score and is off on Tue/Thu; visits on those days lose the strength measurement.	Visits booked Mon/Wed/Fri where possible.	2	N/A	N/A	N/A	3	1	☐ Low ☐ Moderate ☑ High ☐ Critical
SRR-002	2 Jun 2026	Device accountability	Domain: CtQ4 Factor: GLENOS liners kept apart from marketed liners of the same size Source: CIP 7.4; GS-RA-01 item 6	Study and marketed liners share shelf B in the OR store, in similar boxes; a wrong liner may be implanted.	OR nurse reads the label aloud at picking.	2	2	2	3	2	2	☐ Low ☐ Moderate ☑ High ☐ Critical
SRR-003	2 Jun 2026	Device accountability	Domain: CtQ4 Factor: every implanted component reconciled to the participant before case closure Source: CIP 8.3	UDI copied from the OR team without a second check; a missing lot is not caught before case closure.	OR sticker sheet; same-day reconciliation by the coordinator.	2	2	1	3	3	2	☐ Low ☐ Moderate ☑ High ☐ Critical
SRR-004	2 Jun 2026	Safety / clinical event	Domain: CtQ10 Factor: SAE reported to the sponsor within 24 h of site awareness Source: CIP 11.3	Only the coordinator uses the EDC safety form; when she is away nobody can report.	Sub-I listed as backup in the delegation log (not trained on the form).	2	3	N/A	N/A	2	2	☐ Low ☐ Moderate ☑ High ☐ Critical
SRR-005	30 Oct 2026	Imaging / assessment workflow	Domain: CtQ9 Factor: day-1 AP shows liner seating in a reproducible position Source: Imaging manual v2.0, 4.2	Day-1 AP is taken on the ward with the arm in a sling; rotation cannot be controlled and seating is not visible.	None specific; local read of each set (IQL).	3	2	N/A	N/A	3	3	☐ Low ☐ Moderate ☑ High ☐ Critical
SRR-006	30 Oct 2026	Device accountability	Domain: CtQ4 Factor: explanted components quarantined and returned with the lot readable Source: CIP 7.9	No local explant procedure: the liner explanted on 28 Oct stayed unlabeled in theatre for two days.	Coordinator bagged and labeled it on 30 Oct (ad hoc).	2	2	N/A	3	2	3	☐ Low ☐ Moderate ☑ High ☐ Critical
			Domain: Factor: Source:									☐ Low ☐ Moderate ☐ High ☐ Critical

SRR - Site Risk Register

Main register · ownership, action and status

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F Main register · ownership, action and status

Risk status says whether the risk still applies; action status says where the response stands. Implemented is not Effective until a review in G shows the control works.

Risk ID	Risk owner (role)	Review trigger or threshold	Planned action / owner / due	Linked records and evidence	PI / sponsor route and deadline	Risk status	Action status	Last / next review
SRR-001	Physio lead	Any Tue/Thu visit without strength measurement	Train a second physiotherapist · physio lead · 31 Aug	TRE-014-09; PFVR	None needed	☐ Open ☑ Controlled ☐ Accepted ☐ Closed	☐ Not started ☐ In progress ☐ Implemented ☑ Effective ☐ Not effective	31 Aug / 31 Jan
SRR-002	OR nurse lead	Any liner picked from the wrong bin; any near miss	Separate locked, labeled bin · OR nurse lead · 30 Jun	ICDL readiness; IMP check 5	None needed	☐ Open ☑ Controlled ☐ Accepted ☐ Closed	☐ Not started ☐ In progress ☐ Implemented ☑ Effective ☐ Not effective	30 Jul / 31 Jan
SRR-003	Study coordinator	Missing or conflicting UDI; reconciliation after 24 h	Second UDI check, including weekend lists · QA lead · 30 Nov	CAPA-014-002 (reopened); SART-014-005	PI informed 28 Oct; sponsor DD route	☑ Open ☐ Controlled ☐ Accepted ☐ Closed	☐ Not started ☐ In progress ☐ Implemented ☐ Effective ☑ Not effective	31 Oct / 30 Nov
SRR-004	Principal Investigator	Coordinator away for more than one working day	Train the Sub-I on the EDC safety form · coordinator · 15 Nov (was 30 Jun)	TRE-014-12 pending; SART-014-005	POT 3 Nov; sponsor told of late report	☑ Open ☐ Controlled ☐ Accepted ☐ Closed	☐ Not started ☑ In progress ☐ Implemented ☐ Effective ☐ Not effective	31 Oct / 15 Nov
SRR-005	Radiology research tech	Any rotated day-1 image	Ask sponsor to allow day-1 AP in the radiology suite · PI · 30 Nov	IQL-014-10 (R01); Q-IMG-12	Sponsor query 30 Oct	☑ Open ☐ Controlled ☐ Accepted ☐ Closed	☑ Not started ☐ In progress ☐ Implemented ☐ Effective ☐ Not effective	31 Oct / 30 Nov
SRR-006	OR nurse lead	Any explant	Write a local explant procedure · coordinator · 30 Nov	SART-014-005; IMP-014 v1.1 (check 13)	Sponsor RMA route	☑ Open ☐ Controlled ☐ Accepted ☐ Closed	☐ Not started ☑ In progress ☐ Implemented ☐ Effective ☐ Not effective	31 Oct / 30 Nov
						☐ Open ☐ Controlled ☐ Accepted ☐ Closed	☐ Not started ☐ In progress ☐ Implemented ☐ Effective ☐ Not effective	

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Risk review log

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Study / site	SRR ID / revision	Revision effective date	Review log sheet no. / total
GLENOS-01 / 014	SRR-014 / r3	31 Oct 2026	1 / 1

G Risk review log

One line per risk reviewed. Mark whether it is a scheduled review, a trigger or an urgent entry from H. Never overwrite an earlier line.

Date / reviewer	Risk ID	Review type	Trigger and evidence reviewed	Control effectiveness	Current priority and status	Decision / linked record	Next review
30 Jul · Sub-I	SRR-002	☑ Scheduled ☐ Trigger ☐ Urgent (H)	IMC-014-01: bin in use in all 3 cases; no near miss	☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	Moderate · Controlled	Lower priority (I, 30 Jul)	31 Jan
31 Aug · Sub-I	SRR-001	☑ Scheduled ☐ Trigger ☐ Urgent (H)	TRE-014-09: second physio released; no Tue/Thu loss since	☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	Moderate · Controlled	Lower priority (I, 31 Aug)	31 Jan
30 Sep · QA lead	SRR-003	☑ Scheduled ☐ Trigger ☐ Urgent (H)	IMC-014-01: second check done in cases 006-007	☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	Moderate · Controlled	Keep the trigger	31 Oct
28 Oct · coordinator	SRR-003	☐ Scheduled ☐ Trigger ☑ Urgent (H)	SART-014-005: lot missing in ICDL-014-005; second check skipped (weekend, one nurse)	☐ Effective ☐ Partly effective ☑ Not effective ☐ Not assessable	High · Open	Escalate to CAPA (I, 28 Oct)	30 Nov
28 Oct · Sub-I	SRR-004	☐ Scheduled ☑ Trigger ☐ Urgent (H)	SADE reported 27 h after awareness; coordinator away, backup untrained	☐ Effective ☐ Partly effective ☑ Not effective ☐ Not assessable	High · Open	Escalate to POT (I, 3 Nov)	15 Nov
31 Oct · PI	SRR-005	☐ Scheduled ☐ Trigger ☑ Urgent (H)	IQL-014-10: day-1 sets 02 and 07 rotated	☐ Effective ☐ Partly effective ☐ Not effective ☑ Not assessable	High · Open	Sponsor query Q-IMG-12	30 Nov
31 Oct · PI	SRR-006	☐ Scheduled ☐ Trigger ☑ Urgent (H)	Explant left unlabeled for two days	☐ Effective ☐ Partly effective ☐ Not effective ☑ Not assessable	High · Open	Check 13 added to IMP v1.1	30 Nov

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Urgent risks and decisions

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H Emerging or urgent risks

Act on the safety, PI or sponsor route at once; enter the risk in the register afterwards and review it in G. Clinical details stay in the SART or the source record.

No. · date / time	Detected by (role)	Local risk identified	Immediate containment and receiving record	PI notified	Register entry
1 · 28 Oct, 08:40	Coordinator (P-06)	Dislocation (SART-014-005): liner lot missing in ICDL-014-005; no rule for handling the explant.	Lot read from the explant label; explant bagged and labeled 30 Oct; SART-014-005.	☑ Yes ☐ Pending ☐ Not required	☑ New risk ☑ Update ☐ Not required Risk ID: 003; 006
2 · 30 Oct, 10:15	Radiology tech (P-09)	Day-1 ward images rotated twice; liner seating not visible.	Sponsor query Q-IMG-12; IQL-014-10, R01.	☑ Yes ☐ Pending ☐ Not required	☑ New risk ☐ Update ☐ Not required Risk ID: 005
				☐ Yes ☐ Pending ☐ Not required	☐ New risk ☐ Update ☐ Not required Risk ID:

I Status and escalation decisions

Decided and signed here; F is then updated. The PI decides on High and Critical risks and may delegate Low and Moderate.

Lower the priority: only after a review in G shows the controls work; a planned or completed action is not enough. **Accept the residual risk:** no further action is proportionate; record why and keep the trigger. **Close the risk:** the condition no longer applies or the vulnerability was removed with evidence; a closed CAPA alone does not close a risk. **Escalate:** to the CAPA log when a control failed or the problem recurs; to the PI (POT) or a milestone review (MRL) when a decision or a phase gate is affected; to the sponsor when the effect may reach beyond the site or the CIP requires it; to the SART when a safety event may be involved.

Date	Risk ID	Decision	Rule met and evidence	Route / record link	Decided by (PI or delegate)
30 Jul	SRR-002	☑ Lower priority ☐ Accept residual ☐ Close ☐ Escalate	Review G 30 Jul shows the bin works; re-rated Device 3 to 2	IMC-014-01	A. Morgan (PI)
31 Aug	SRR-001	☑ Lower priority ☐ Accept residual ☐ Close ☐ Escalate	Second physio released; Data 3 to 2	TRE-014-09	A. Morgan (PI)
28 Oct	SRR-003	☐ Lower priority ☐ Accept residual ☐ Close ☑ Escalate	Control failed after a CAPA: recurrence	CAPA-014-002 reopened	R. Teixeira (backup PI)
3 Nov	SRR-004	☐ Lower priority ☐ Accept residual ☐ Close ☑ Escalate	Backup never trained; SADE reported late	POT 3 Nov; sponsor letter	A. Morgan (PI)

J Register review certification

At each review of the whole register. The site sets how often this happens.

Review date	31 Oct 2026	**Review frequency set by the site**	Quarterly, and after any SADE
Open High / Critical risks	SRR-003, SRR-004, SRR-005, SRR-006		
Accepted residual risks	None		
Next scheduled SRR review	31 Jan 2027, or earlier on a trigger	**Linked POT / MRL review ID**	POT 3 Nov 2026

Overall assessment · what changed since the last review, where the site is most exposed, what needs attention next

October changed the picture. Two of our controls failed in the same case (SART-014-005): the second UDI check was skipped on a weekend with one nurse, and the backup for safety reporting had never been trained. Both rest on single people. The day-1 image problem and the missing explant procedure were invisible until the dislocation.

Most exposed: weekend and holiday cover for OR and safety tasks. Next: train the Sub-I on the EDC safety form by 15 Nov, extend the second check to weekend lists, and get the sponsor's answer on day-1 imaging before the next inclusion.

I confirm that the risks in this register were reviewed against current evidence and the controls in place, and that actions, triggers and communications are assigned or linked. This does not mean the risks are gone.

Register owner

Name / role: Maria Silva / Study Coordinator

Signature / date:

M. Silva 31 Oct 2026

Principal Investigator

Name: Dr Alex Morgan

Signature / date:

A. Morgan 3 Nov 2026